

Question 5. In the code cell below, create a visualization that will help us determine if there is an association between birth rate and death rate during this time interval. It may be helpful to create an intermediate table here. **(4 Points)**

Things to consider:

- What type of chart will help us illustrate an association between 2 variables?
- How can you manipulate a certain table to help generate your chart?
- Check out the Recommended Reading for this homework!

```
In [ ]: # In this cell, use birth_rates and death_rates to generate your visualization
        birth_rates = pop.column('BIRTHS') / pop.column('2015')
        death_rates = pop.column('DEATHS') / pop.column('2015')
        table = Table().with_columns('Birth Rates', birth_rates, 'Death Rates', death_rates)
        table.scatter('Birth Rates', 'Death Rates')
```


Question 1. Produce a histogram that visualizes the distributions of all ride times in Boston using the given bins in `equal_bins`. (4 Points)

```
In [ ]: equal_bins = np.arange(0, 120, 5)
        boston.hist('ride time', bins=equal_bins)
```


Question 2. Now, produce a histogram that visualizes the distribution of all ride times in Manila using the given bins. (4 Points)

```
In [ ]: equal_bins = np.arange(0, 120, 5)
        manila.hist('ride time', bins=equal_bins)

        # Don't delete the following line!
        plots.ylim(0, 0.05);
```


Question 6. Identify one difference between the histograms, in terms of the statistical properties. Can you comment on the average and/or skew of each histogram? **(4 Points)**

Hint: The best way to do this is to compare the two histograms (from 3.1 and 3.2) visually.

The mean ride time for Boston is lower than the mean ride time in Manila because the boston values are tightly clustered to the left side of the graph, while Manila has more values across longer periods of times on the right side of the graph.

Question 7. Why is your solution in Question 6 the case? Based on one of the following two readings, why are the distributions for Boston and Manila different? **(4 Points)**

- [the Boston climate summary](#)
- [this article on Manila traffic](#)

Hint: Try thinking about external factors of the two cities that may be causing the difference! There may be multiple different factors that come into play.

Manila has longer drive times because of poor city infrastructure that can cause drive times to be unpredictable.

Question 2. State at least one reason why you chose the histogram from Question 1. **Make sure to clearly indicate which histogram you selected** (ex: “I chose histogram A because ...”). **(5 Points)**

I chose histogram 3 because its distribution of numbers on the x axis correlated to the ones on the x axis of the scatter chart, and that the mode seemed to be around 0.5 for both.

Question 4. State at least one reason why you chose the histogram from Question 3. **Make sure to clearly indicate which histogram you selected** (ex: “I chose histogram A because ...”). **(5 Points)**

I chose histogram 2 because there is no y values for -1 to 1 on the y axis in both charts.

